Stanley Chukwuebuka Okonkwo

MSc, Ecology and Nature Management — RUDN University, Moscow (2026)

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EPFL Doctoral School

Selection Committee — PhD Position Ref. 2222

EPFL Valais, Sion, Switzerland

Re: Application for PhD Position: Forest Treeline Management in the Swiss Alps

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Dear Selection Committee,

I am applying for the PhD position in forest treeline management at EPFL Valais in Sion. I recently completed an MSc in Ecology and Nature Management at RUDN University in Moscow and am seeking a PhD position that will allow me to develop my remote sensing and ecological modelling capabilities within a rigorous research environment. The methodological combination required by this position, namely high-resolution spatial analysis, ecological predictive modelling, and applied land management, maps closely onto the foundation I have built through two degrees of research in the earth and environmental sciences.

My MSc thesis examines the ecological impacts of urbanisation on green spaces and drainage patterns in Lagos State, Nigeria, using multi-temporal Landsat 8/9 and Sentinel-2 satellite imagery processed in Google Earth Engine and ArcGIS. The analytical core of the work is change detection: tracking vegetation cover loss and drainage network disruption over time using NDVI time-series analysis and land surface temperature mapping. I designed and executed the entire remote sensing pipeline independently, from image acquisition and preprocessing through classification, accuracy assessment, and spatial analysis at the state level. This is the same class of methodological work that mapping treeline dynamics from high-resolution satellite data requires, applied at a different spatial scale and ecological context.

Alongside my thesis I developed EcoSim, a web-based ecological simulation platform that models degradation and restoration dynamics for five West and Central African biomes using a quantitative state model and stochastic event system. Building this required me to translate ecological processes, including vegetation succession, soil hydration dynamics, heat island effects, and biodiversity recovery, into computational models that behave in ecologically plausible ways. I am proficient in Python and have experience with scientific data processing, geospatial libraries, and the integration of observational datasets into simulation frameworks. Developing deep learning models for treeline mapping and designing predictive models for climate scenarios would extend this existing capability into new methodological territory, which I regard as an opportunity.

I want to be direct about the disciplinary shift this position represents. My research to date has focused on tropical urban ecosystems in West Africa. Forest treeline dynamics in the Swiss Alps is a different system and a different set of ecological questions. What it shares with my existing work is the central methodological challenge: using satellite-derived spatial data to understand how ecosystems are changing under climatic and human pressure, and what those changes mean for the landscapes they affect. The treeline is one of the most sensitive indicators of climate-driven ecological change available to researchers. Working on it at EPFL, with the tools and supervision of one of the leading environmental science institutions in the world, would represent a level of scientific development I could not achieve elsewhere.

The participatory and applied dimension of the project is also important to me. The position description mentions co-creation workshops with forest professionals and the design of management tools for practitioners. My experience building accessible, web-based environmental simulation tools has taught me that the gap between scientific research and practical application is as much a design problem as a knowledge problem. I have thought seriously about how to make ecological analysis usable for people who are not scientists, and I would actively contribute that perspective to the outreach and co-design components of this project.

I completed my MSc in June 2026 and am available to begin on 1 September 2026 as advertised. I hold a BSc in Geological Sciences from Nnamdi Azikiwe University, Nigeria (2019), have two peer-reviewed publications in environmental and geoscience journals, and presented my MSc research at the Lomonosov International Conference at Moscow State University in April 2025. Three referees are available: my MSc supervisor Professor Pinaev Vladimir Evgenievich at RUDN University, my BSc thesis supervisor Dr. Nwokeabia Charity Nkiru at Nnamdi Azikiwe University, and a further academic contact from my research network.

I would welcome the opportunity to discuss my application further.

Yours sincerely,

Stanley Chukwuebuka Okonkwo
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